Pass-Through, Welfare, and Incidence under Product Returns in Perfect Competition

Draft notes

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1 Proposal

About one in five products sold online is returned. Pass-through, tax incidence, and the efficiency cost of taxation are well understood when purchase settles who keeps the good. In that classical setting, the demand and supply curves are sufficient for consumer surplus, producer surplus, and deadweight loss. With returns, allocation is decided in two stages. Buyers first choose whether to purchase, then whether to keep or return after receiving the item. Never buying and buying then returning leave the same final allocation—the consumer does not keep the good—yet their welfare costs differ. A returned purchase incurs shipping, hassle, and handling that a non-purchase avoids, so demand and supply alone omit an economically relevant margin.

What objects then replace the classical demand and supply curves, and how misleading is an analysis that ignores returns? We study these questions under perfect competition with product returns and seller handling costs. We characterize pass-through, incidence, and welfare losses with returns, compare them to the no-return benchmark, and identify where omitting the return margin changes conclusions.

Consumer surplus depends only on two aggregates: gross demand and the return rate. Heterogeneity in tastes, match values, and return costs enters welfare solely through these objects. Producer surplus retains the usual geometry—area to the left of the supply curve—but evaluated at net revenue rather than at list price. Net revenue is list price minus expected refunds and handling.

Because consumers pay the list price while firms respond to net revenue, pass-through from cost shocks into list prices is governed by how net revenue moves with the list price. Tax neutrality fails for the same reason. The keep-or-return decision depends on the refund price, which differs according to which party remits the tax. An analyst who ignores returns and applies the classical formulas therefore misstates incidence and deadweight loss whenever the return margin is active.

2 Setting

We begin with the market primitives: purchase and return behavior on the consumer side, and competitive supply with return handling on the firm side. A homogeneous product is sold at list price p. Consumers indexed by i differ in return hassle κ_i, the disutility from returning the product, and in match distribution G_i over post-purchase match quality u. After purchase, type i draws $u \sim G_i$ and keeps the item whenever $u - p \geq -\kappa_i$, earning surplus $u - p$, and returns it otherwise, earning $-\kappa_i$.

Buying at price p yields expected surplus

$$s_i(p) = \mathbb{E}_{u\sim G_i}\big[\max(u-p,\,-\kappa_i)\big].$$

On the purchase margin, all consumer-side heterogeneity is summarized by willingness to pay v_i, defined by $s_i(v_i)=0$, so type i buys if and only if $p \le v_i$. Across types, willingness to pay has distribution $F(v)=\Pr(v_i \le v)$ with density f, so gross demand is $D(p)=1-F(p)$. Of the units sold at price p, a share $r(p)$ are returned.

To understand how $r(p)$ changes with price, write type i's return probability as $\rho_i(p)=\Pr(u < p-\kappa_i)$. Willingness to pay is sufficient for purchase but not for returns: types with the same v_i may still differ in (κ_i, G_i). Averaging within WTP cells,

$$\rho(v,p) \equiv \mathbb{E}\big[\rho_i(p) \;\big|\; v_i = v\big],$$

so the aggregate return rate is the truncated mean

$$r(p) = \mathbb{E}\big[\rho(v_i,p) \;\big|\; v_i \ge p\big].$$

Differentiating yields

$$r'(p) = \mathbb{E}\big[\partial\rho(v_i,p)/\partial p \;\big|\; v_i \ge p\big] + \frac{f(p)}{D(p)}\big(r(p)-\rho(p,p)\big),$$

where $\rho(p,p)$ is the average return probability among types at the purchase margin $v_i = p$.[1] The first term is behavioral: a higher keep threshold raises return probabilities among those who still buy. The second is selection: a higher price drops cutoff types at the purchase margin, and that term would be negative if those cutoff types return more than the average purchaser.[2] Behavioral and selection effects can therefore pull in opposite directions, and we do not impose a sign on $r'(p)$.

The product market is perfectly competitive. Producer j takes p as given, chooses output at cost $C_j(q_j)$, and incurs a net return handling cost h on each returned unit, equal to gross handling cost minus the scrap value of the returned product. This net cost may be negative, but we maintain $p+h>0$, which holds whenever scrap value does not exceed list price.

3 Consumer side

We turn first to consumer surplus and show that only the aggregates D and r enter. Write $s(v,p)$ for expected surplus at list price p among types with willingness to pay v, so $s(v,v)=0$. Consumer surplus integrates surplus over all types who buy,

$$CS(p) = \int_p^\infty s(v,p)\,dF(v). \tag{1}$$

[1] With $\rho(v,p)$ as defined in the text, $r(p) = D(p)^{-1}\int_p^\infty \rho(v,p)\,dF(v)$. Differentiating and using $D'(p) = -f(p)$ gives

$$r'(p) = D(p)^{-1}\int_p^\infty \partial\rho(v,p)/\partial p\,dF(v) + \frac{f(p)}{D(p)}\big(r(p)-\rho(p,p)\big),$$

which is the display in the text.

[2] Selection need not be negative for arbitrary (κ_i, G_i). It is signed under an additive match specification $u = v_i + \varepsilon$ with shock ε of any distribution independent of willingness to pay, and with return hassle likewise independent of v: then $\rho(v,p)$ decreases in v, so $\rho(p,p) \ge r(p)$ and the selection term is non-positive.

Differentiating in p gives an extensive-margin term from types entering or leaving the purchase margin and an intensive-margin term from lower surplus among inframarginal buyers,

$$CS'(p) = -s(p,p)\, f(p) + \int_p^{\infty} s_p(v,p)\, dF(v),$$

where $s_p(v,p) \equiv \partial s(v,p)/\partial p$. On the extensive margin, $s(p,p) = 0$, so entry and exit contribute nothing at first order. On the intensive margin, $s_p(v,p)$ equals minus the keep probability, because a higher price harms the consumer only when she keeps the purchase,

$$s_p(v,p) = -\Pr(u \geq p - \kappa_i).$$

Aggregating over buyers who purchase then gives

$$CS'(p) = \int_p^{\infty} s_p(v,p)\, dF(v) = -D(p)\big(1 - r(p)\big),$$

by the definitions of $D(p)$ and $r(p)$. A small price change therefore satisfies

$$dCS = -D(p)\big(1 - r(p)\big)\, dp. \tag{2}$$

The extensive- and intensive-margin decomposition makes the local formula transparent. Consumer surplus changes depend only on the aggregates $D(p)$ and $r(p)$, both at each price and integrated along any path,

$$\Delta CS = -\int_{p_0}^{p_1} D(p)\big(1 - r(p)\big)\, dp.$$

Micro heterogeneity in price responses can be rich. Price p enters each $s_i(p)$ nonlinearly and can shift participation, return decisions, and inframarginal surplus differently across types. Yet once (D, r) are given, this microstructure is irrelevant for dCS and ΔCS. Two economies with the same paths of $D(p)$ and $r(p)$ deliver the same consumer surplus loss, whether high-v_i types adjust return behavior more steeply than low-v_i types or not. Only kept units enter at first order, through $1 - r(p)$, not gross purchases $D(p)$ alone.

Let $\bar{p}$ denote the highest willingness to pay in the population, so $D(\bar{p}) = 0$ and $CS(\bar{p}) = 0$. Integrating the local formula then rewrites consumer surplus as area under kept volume,

$$CS(p) = \int_p^{\bar{p}} D(z)\big(1 - r(z)\big)\, dz. \tag{3}$$

4 Supply side

On the firm side, returns make list price an imperfect summary of producer payoffs. A gross sale brings revenue p if the consumer keeps the item and costs refund p plus net handling h if the item is returned. Among gross sales at list price p, share $r(p)$ are returned, so expected revenue per gross unit—the *effective marginal revenue*—is

$$\mu(p,h) \equiv p - r(p)(p + h). \tag{4}$$

Under perfect competition, firm j takes p as given and therefore faces a fixed number μ in its supply decision. At equilibrium, $\mu = \mu(p,h)$.

Given μ, firm j chooses output to solve

$$\max_{q_j \geq 0} \ \mu\, q_j - C_j(q_j), \qquad \text{FOC: } C_j'(q_j) = \mu \ \ (\text{if } q_j > 0),$$

with $C_j(0) = 0$. Write $q_j(\mu)$ for firm j's supply, set to zero when μ is below its minimum marginal cost.

Summing across firms defines market supply and industry profit,

$$S(\mu) \equiv \sum_j q_j(\mu), \qquad PS(\mu) \equiv \sum_j \big[\mu\, q_j(\mu) - C_j\big(q_j(\mu)\big)\big]. \tag{5}$$

Which firms are active at each μ determines both the market quantity and total profit.

Differentiating $PS(\mu)$ with respect to μ gives, for each firm,

$$\frac{d}{d\mu}\Big[\mu\, q_j - C_j(q_j)\Big] = q_j + q_j'(\mu)\, \big[\mu - C_j'(q_j)\big].$$

Active firms satisfy $C_j'(q_j) = \mu$ at their chosen output, so the adjustment term $q_j'(\mu)\,[\mu - C_j'(q_j)]$ is zero for each firm—equivalently, the envelope theorem applied to each firm's profit problem gives $d\pi_j/d\mu = q_j(\mu)$. Summing over firms,

$$PS'(\mu) = \sum_j q_j(\mu) = S(\mu).$$

With no production at $\mu = 0$, it follows that industry profit equals the area to the left of the market supply curve,

$$PS(\mu) = \int_0^{\mu} S(z)\, dz. \tag{6}$$

At equilibrium $\mu = \mu(p, h)$, $PS = \int_0^{\mu(p,h)} S(z)\, dz$.

5 Failure of tax neutrality

With firms responding to μ rather than list price, a natural question is whether remittance of a unit tax remains neutral. Weyl and Fabinger (2013) show that without returns, buyer and seller remittance deliver the same equilibrium. With returns, we ask whether that equivalence survives by writing the two clearing conditions at a common checkout price p—the total payment at the register—and a per-unit tax τ. Fix p and τ. Seller remittance clears at

$$D(p) = S\big(p - \tau - r(p)(p + h)\big), \tag{7}$$

because both the return rate and per-return cost $p + h$ are evaluated at checkout price p. Buyer remittance clears at

$$D(p) = S\big((p - \tau) - r(p - \tau)(p - \tau + h)\big), \tag{8}$$

because the seller posts list price $p - \tau$, so returns and refund cost are evaluated there. The two conditions define the same equilibrium only if the arguments of $S(\cdot)$ coincide. Canceling $p - \tau$ yields

$$r(p)(p + h) = r(p - \tau)(p - \tau + h). \tag{9}$$

Result 1 (Failure of tax neutrality with returns). *Seller and buyer taxes define the same equilibrium if and only if* $r(p)(p+h) = r(p-\tau)(p-\tau+h)$.[3]

Tax neutrality breaks down with product returns because the refunded amount—and therefore the keep-or-return decision—depends on which side of the market remits the tax. Under seller remittance, the checkout price (which includes the tax) is refunded, so consumers decide whether to return based on that tax-inclusive price. Under buyer remittance, the seller posts a pre-tax list price, only that list price is refunded, and the return decision is based on this lower, tax-exclusive price. Even though the total checkout price and the tax are identical in both regimes, the return rate and the seller's refund liability per return are evaluated at different prices (p versus $p-\tau$).

The direction in which the return rate changes is ambiguous—a higher price can make each individual more likely to return but can also select consumers with stronger purchase intentions who are less likely to return—so $r(p)$ may be above or below $r(p-\tau)$. Regardless of the direction, the seller's effective marginal revenue generally differs, so the two tax regimes produce distinct market-clearing conditions and are not equivalent. The fundamental cause is that the purchase decision is always based on the checkout price, while the return decision uses the tax-inclusive price under seller remittance and the tax-exclusive price under buyer remittance, creating a wedge that alters equilibrium.

6 Pass-through

Equilibrium satisfies $D(p) = S(\mu(p,h))$. Write $Q \equiv D(p)$. We characterize pass-through into list price for three shocks: a uniform one-dollar parallel shift in every firm's marginal production cost (indexed by c), a one-dollar increase in handling cost h, and a unit tax under seller and buyer remittance. Production cost shifts market supply at a given effective margin μ, while handling lowers $\mu(p,h)$ at a fixed list price p.

For production and handling, write the clearing condition as

$$m(p,x) \equiv D(p) - S(\mu(p,x),x) = 0, \qquad x \in \{c,h\}. \tag{10}$$

Differentiating while keeping $m = 0$ defines pass-through $\rho_x \equiv dp/dx$ by

$$\rho_x = -\frac{m_x}{m_p}, \tag{11}$$

and expanding $m = D - S$ gives

$$m_p = D'(p) - S'(\mu)\,\mu_p, \qquad\qquad m_x = -S'(\mu)\,\mu_x - S_x, \tag{12}$$

hence

$$\rho_x = \frac{-S'(\mu)\,\mu_x - S_x}{S'(\mu)\,\mu_p - D'(p)}. \tag{13}$$

The two shocks differ in which margin moves at fixed p or fixed μ,

$$\mu_c = 0, \qquad S_c = -S'(\mu), \qquad \mu_h = -r(p), \qquad S_h = 0. \tag{14}$$

The production term is $S_c = -S'(\mu)$ because a one-dollar parallel rise in every firm's marginal cost is equivalent, at fixed μ, to evaluating market supply at $\mu - 1$. Substituting into (13) yields

[3] Neutrality requires the return wedge to be the same at checkout price p and at posted list price $p-\tau$. That holds for all taxes only if $r(p)(p+h)$ is constant in posted list price, which requires $r(p) = C/(p+h)$ for some constant C; without that functional form on r, the two clearing conditions need not define the same equilibrium.

Result 2 (Pass-through of production and handling costs)**.**

$$\frac{\rho_h}{\rho_c} = r(p), \tag{15}$$

with pass-through into list price

$$\rho_c = \frac{S'(\mu)}{S'(\mu)\,\mu_p - D'(p)}, \qquad \rho_h = r(p)\,\rho_c. \tag{16}$$

Handling moves μ by only $r(p)$ per dollar relative to a unit production shift, so $\rho_h/\rho_c = r(p)$. When $r \equiv 0$, handling does not move μ and ρ_c coincides with the textbook competitive pass-through formula.

Turning to the unit tax, because of the failure of tax neutrality we write p^s for checkout price under seller remittance and p^b for checkout price under buyer remittance; let $\rho^s_\tau \equiv dp^s/d\tau$ and $\rho^b_\tau \equiv dp^b/d\tau$. For a local change at $\tau = 0$, tax pass-through is (13) with $S_\tau = 0$: the denominator uses the same μ_p as above, and the only difference across remittance regimes is μ^s_τ versus μ^b_τ at fixed checkout price:

$$\rho^s_\tau = \frac{-S'(\mu)\,\mu^s_\tau}{S'(\mu)\,\mu_p - D'(p^s)}, \qquad \rho^b_\tau = \frac{-S'(\mu)\,\mu^b_\tau}{S'(\mu)\,\mu_p - D'(p^b)}. \tag{17}$$

At $\tau = 0$,

$$\mu^s_\tau = -1, \qquad \mu^b_\tau = r(p^b) + (p^b + h)\,r'(p^b) - 1. \tag{18}$$

Under seller remittance, $\mu^s_\tau = -1$ at fixed checkout price, so $\rho^s_\tau = \rho_c$. Under buyer remittance, μ^b_τ differs, so tax pass-through generally differs from ρ_c.

6.1 Elasticity representation

The same pass-through formulas can be rewritten in elasticities, which makes the role of μ_p transparent. Write $\mu_p \equiv d\mu/dp$ for the response of effective revenue to list price, and define demand and supply elasticities with respect to list price as usual by

$$\varepsilon_D(p) \equiv \frac{pD'(p)}{D(p)} < 0, \qquad \varepsilon_S(p) \equiv \frac{dS(\mu(p))}{dp}\,\frac{p}{S(\mu(p))} = S'(\mu)\,\mu_p\,\frac{p}{S(\mu)} > 0.$$

Here $\varepsilon_S(p)$ is the observed elasticity of market supply with respect to the clearing price.

Without returns, supply is a function of list price, so $S' = dS/dp$. A one-dollar cost increase cuts supply by S' units at the original price, creating an excess-demand gap of that size. Raising list price by one dollar closes the gap from both sides. Demand falls by $|D'|$ and supply recovers by S'. Closing a gap of size S' therefore requires a price rise of $S'/(S'+|D'|)$, so standard pass-through equals $\varepsilon_S/(-\varepsilon_D + \varepsilon_S)$.

With returns, supply is a function of effective revenue, so $S'(\mu) = dS/d\mu$ rather than dS/dp. A one-dollar production-cost increase is equivalent to lowering μ by one dollar, so at fixed list price supply falls by $S'(\mu)$ units, again creating an excess-demand gap of that size. Raising list price by one dollar closes the gap at rate $S'(\mu)\,\mu_p - D'(p)$, so

$$\rho_c = \frac{S'(\mu)}{S'(\mu)\,\mu_p - D'(p)}.$$

The cost shock lowers μ by one dollar, while a dollar of list price raises μ by μ_p. A higher μ_p means a one-dollar price increase restores more effective revenue—and therefore more quantity supplied—than the one-dollar cost shock removes, so a smaller list-price movement closes the excess-demand gap. Measuring supply against list price through ε_S puts the factor $1/\mu_p$ in the numerator,

$$\rho_c = \frac{\varepsilon_S(p)/\mu_p}{-\varepsilon_D(p) + \varepsilon_S(p)}. \tag{19}$$

Pass-through is therefore dampened relative to the no-return benchmark when $\mu_p > 1$ and amplified when $\mu_p < 1$. If returns are absent, $r \equiv 0$ forces $\mu_p = 1$, and (19) collapses to the standard formula.

Whether μ_p exceeds one depends on how steeply the return rate falls with price. Differentiating $\mu = p - r(p)(p+h)$ gives

$$\mu_p = 1 - r(p) - r'(p)(p+h).$$

Write $\varepsilon_r(p) \equiv p\, r'(p)/r(p)$ for the elasticity of the return rate in list price. Then $\mu_p > 1$ if and only if $\varepsilon_r(p) < -p/(p+h)$. That is, selection must outweigh the behavioral rise in individual return probabilities by enough that aggregate r falls steeply in p. When instead the behavioral response dominates, ε_r is less negative (or positive) and $\mu_p < 1$. This comparison of μ_p with the classical benchmark $\mu_p = 1$ implies a directional bias for an analyst who ignores returns.

Result 3. *Relative to the correct returns-adjusted ρ_c, the classical no-return pass-through formula overestimates pass-through when $\varepsilon_r(p) < -p/(p+h)$ and underestimates it when the behavioral response dominates selection.*

The Result is therefore an easy test for the direction of the bias from observables—the return-rate elasticity and the size of handling relative to list price. Intuitively, effective revenue puts weight $p+h$ on each returned unit. When selection lowers the return rate as price rises, a higher handling cost magnifies how much that decline in returns improves μ, so μ_p exceeds one more readily and the classical formula overstates pass-through. When handling is small, the return margin matters less for μ, and the level effect of returns (the $-r$ term in μ_p) is more likely to leave $\mu_p < 1$, so the classical formula understates pass-through.

7 Incidence

Pass-through into list price determines the welfare loss from each cost shifter and how that loss is split between consumers and producers. Throughout, $Q \equiv D(p)$ and $r \equiv r(p)$ denote the equilibrium gross quantity and the return rate. From the consumer-side analysis, a small increase in any shifter x changes consumer surplus by

$$\frac{dCS}{dx} = -Q\,(1-r)\,\rho_x, \tag{20}$$

where $\rho_x \equiv dp/dx$ is the pass-through of the shock into the list price.

Producer surplus is less immediate because a shock may shift the supply schedule directly as well as indirectly through the effective marginal revenue μ. Let $S(\mu, x)$ be the market supply curve, where the second argument allows for a direct effect of x on firms' costs, and write producer surplus as $PS(\mu, x) = \int_0^{\mu} S(z,x)\,dz$. Differentiating with respect to x,

$$\frac{dPS}{dx} = S(\mu, x)\,\frac{d\mu}{dx} + \int_0^{\mu} \frac{\partial S(z,x)}{\partial x}\,dz.$$

Using the equilibrium condition $S(\mu, x) = Q$ and denoting $\Sigma_x \equiv \int_0^{\mu} (\partial S(z, x)/\partial x)\, dz$, we obtain

$$\frac{dPS}{dx} = Q\,\frac{d\mu}{dx} + \Sigma_x = Q\big(\mu_p\, \rho_x + \mu_x\big) + \Sigma_x, \tag{21}$$

with $\mu_p \equiv \partial\mu/\partial p$ and $\mu_x \equiv \partial\mu/\partial x$ evaluated at fixed list price.

The term Σ_x captures the direct shift of the supply curve holding μ constant. A uniform, parallel increase in every firm's marginal production cost by one dollar ($x = c$) moves the supply curve inward according to $\partial S(\mu, c)/\partial c = -S'(\mu)$, so $\Sigma_c = \int_0^{\mu} (-S'(z))\, dz = -S(\mu) = -Q$. For a handling-cost shock h or a per-unit tax (whether seller- or buyer-remitted), the supply schedule at a given μ is unchanged, hence $\Sigma_h = \Sigma_{\tau_s} = \Sigma_{\tau_b} = 0$. The partial derivatives μ_x are obtained directly from the definition of effective marginal revenue: $\mu_c = 0$, $\mu_h = -r$, $\mu_{\tau_s} = -1$, and, for the buyer-remitted tax evaluated at $\tau = 0$, $\mu_{\tau_b} = r + (p + h)r' - 1$ as derived in Section 5.

Substituting these values together with the pass-through expressions $\rho_h = r\rho_c$ and $\rho_\tau^s = \rho_c$ into (21) yields the producer-surplus responses:

$$\frac{dPS}{dc} = Q\big(\mu_p\, \rho_c - 1\big), \tag{22}$$
$$\frac{dPS}{dh} = Q\big(\mu_p\, \rho_h - r\big), \tag{23}$$
$$\frac{dPS}{d\tau_s} = Q\big(\mu_p\, \rho_c - 1\big), \tag{24}$$
$$\frac{dPS}{d\tau_b} = Q\big(\mu_p\, \rho_\tau^b + \mu_{\tau_b}\big). \tag{25}$$

The incidence ratio $I(x) \equiv |dCS/dx| \,/\, |dPS/dx|$ measures how the total loss is divided. For production cost, handling cost, and the seller-remitted tax, the ratio simplifies to the same expression:

$$I(c) = I(h) = I(\tau_s) = \frac{(1 - r)\, \rho_c}{1 - \mu_p\, \rho_c}. \tag{26}$$

For the buyer-remitted tax, the ratio is

$$I(\tau_b) = \frac{(1 - r)\, \rho_\tau^b}{-\big(\mu_p\, \rho_\tau^b + \mu_{\tau_b}\big)}, \qquad \mu_{\tau_b} = r + (p + h)r' - 1. \tag{27}$$

Equation (26) generalises the standard competitive-market formula $I = \rho/(1 - \rho)$ (Weyl and Fabinger, 2013). Consumer loss is scaled by the keep probability $1 - r$, and the producer-side denominator involves the effective-revenue pass-through $\mu_p\rho_c$ rather than the list-price pass-through alone. Production cost, handling cost, and a seller-remitted tax all generate the same incidence split because none of them changes the price at which consumers decide whether to return the product; they shift equilibrium exclusively through the effective marginal revenue μ. The buyer-remitted tax, in contrast, alters the refund price to the pre-tax list price, so the return rate and the per-return refund liability differ. This breaks the symmetry, giving an incidence ratio that departs from (26)—a welfare counterpart to the tax-neutrality failure established earlier. When returns are absent ($r = 0$, $\mu_p = 1$, $\rho_\tau^b = \rho_c$), all four ratios collapse to $\rho_c/(1 - \rho_c)$, restoring the familiar equivalence.